

CALLAGHAN VALLEY, BC, Canada

WMO: 716880

Lat: 50.144N

Lon: 123.111W

Elev: 884

StdP: 91.15

Time Zone: -8.00 (NAP)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-14.3	-12.0	-19.2	0.8	-12.1	-15.6	1.1	-9.9	3.3	1.3	2.9	-0.1	0.9	80	0.597

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.4	28.7	15.2	26.5	14.6	24.3	14.0	16.5	25.4	15.5	24.1	14.7	22.4	1.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.4	10.7	17.8	12.5	10.0	16.7	11.7	9.5	15.8	49.1	25.8	46.4	24.0	43.8	22.6	19.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 3.2	2.7	2.4	DB	-17.4	31.9	3.1	1.8	-19.7	33.2	-21.5	34.2	-23.3	35.3	-25.5	36.6	(4)
(5)			WB	-17.8	17.8	3.1	1.2	-20.0	18.6	-21.8	19.4	-23.6	20.0	-25.9	20.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.8	-1.1	-1.1	1.0	3.7	8.4	11.1	15.4	15.7	11.6	6.0	0.7	-2.5		
	DBStd	7.21	3.46	4.01	3.06	2.85	3.42	3.62	3.63	3.58	3.96	3.10	3.63	3.60		
	HDD10.0	2048	345	311	279	191	74	26	2	1	24	131	278	386		
	HDD18.3	4616	603	544	537	439	309	219	107	98	204	382	528	644		
	CDD10.0	512	0	0	0	2	24	59	168	178	74	7	0	0		
	CDD18.3	38	0	0	0	0	0	3	15	17	3	0	0	0		
	CDH23.3	657	0	0	0	0	9	73	262	249	63	0	0	0		
	CDH26.7	170	0	0	0	0	1	19	73	64	12	0	0	0		
(14) Wind	WSAvg	0.9	0.6	0.8	0.9	1.1	1.1	1.1	1.1	1.0	0.9	0.8	0.7	0.6		
(15) Precipitation	PrecAvg	918	104	66	75	55	51	69	50	52	61	96	128	109		
	PrecMax	1177	215	148	131	102	99	137	119	136	128	178	265	161		
	PrecMin	718	56	2	15	24	21	34	10	12	7	9	51	50		
	PrecStd	116	36	32	29	20	20	24	25	28	33	40	53	29		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.0	10.2	13.9	18.9	24.4	29.5	31.8	30.8	28.4	19.6	10.1	6.2		
		MCWB	4.5	3.8	6.2	9.3	12.1	14.8	16.4	16.0	14.7	11.2	8.4	5.5		
	2%	DB	4.4	6.7	9.8	14.1	21.5	25.4	28.6	28.6	25.1	15.6	7.8	3.6		
		MCWB	3.0	3.0	4.7	6.4	11.4	14.0	15.6	15.2	13.8	9.8	6.6	3.1		
	5%	DB	3.0	4.7	6.9	11.0	18.6	21.6	26.3	26.2	21.6	12.3	6.4	2.3		
		MCWB	2.3	2.9	3.5	5.0	10.2	12.8	14.8	14.8	12.9	8.9	5.6	1.9		
	10%	DB	1.9	3.1	5.0	8.4	15.6	18.4	23.9	23.7	17.8	10.2	5.2	1.3		
		MCWB	1.4	1.9	2.7	4.0	9.2	11.5	14.3	14.2	11.8	7.8	4.6	1.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.7	5.3	7.7	10.0	13.9	16.1	18.0	17.2	16.2	12.6	9.3	5.7		
		MCDB	6.0	6.9	11.9	17.9	20.7	25.8	28.1	25.9	24.9	16.0	10.0	6.0		
	2%	WB	3.2	4.2	5.5	7.6	12.0	14.6	16.7	16.3	14.8	11.0	6.9	3.2		
		MCDB	4.0	5.9	8.5	12.8	19.8	23.4	25.7	25.1	21.4	14.6	7.4	3.5		
	5%	WB	2.3	3.0	4.0	5.7	10.8	13.2	15.6	15.5	13.7	9.5	5.7	2.0		
		MCDB	2.9	4.2	6.3	9.5	17.2	20.5	24.3	24.0	19.5	11.9	6.3	2.2		
	10%	WB	1.5	1.9	2.9	4.5	9.5	11.8	14.6	14.7	12.4	8.2	4.6	1.0		
		MCDB	1.9	2.9	4.6	7.5	14.9	17.4	22.4	22.3	17.2	9.9	5.1	1.2		
(35) Mean Daily Temperature Range	5% DB	MDBR	5.2	7.1	7.4	8.7	11.5	10.3	11.7	11.4	9.6	7.2	4.9	5.0		
		MCDBR	5.7	8.9	11.2	14.1	17.0	16.7	16.7	16.2	15.5	11.7	6.5	5.2		
	5% WB	MCWBR	4.3	5.5	6.2	7.0	8.1	7.4	6.3	5.6	6.4	6.6	4.9	4.5		
		MCWBR	5.3	7.1	9.4	12.0	14.7	14.9	14.8	14.1	13.1	10.4	5.6	4.6		
(40) Clear-Sky Solar Irradiance	taub	taub	0.269	0.271	0.291	0.329	0.343	0.348	0.340	0.336	0.312	0.308	0.276	0.261		
		taud	2.264	2.292	2.244	2.151	2.243	2.350	2.402	2.442	2.517	2.489	2.373	2.265		
	Edn at Noon	Edn at Noon	771	875	916	911	912	908	910	899	891	827	764	721		
		Edh at Noon	69	87	110	136	130	119	111	101	84	72	62	60		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.86	1.72	2.52	4.04	5.31	5.29	5.92	5.15	3.54	1.97	0.96	0.69		
	RadStd	RadStd	0.15	0.26	0.40	0.50	0.51	0.66	0.59	0.39	0.31	0.32	0.19	0.12		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (51 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page